Human escalation workflows

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Ask who can decide, what they need and what remains pending.

Teach scoped reviewer routes, accurate reported facts and explicit communication limits.

Keep one habit: Model incomplete handoff → assign correct role → inspect unreachable case → assess deadline reasoning → record unresolved fallback.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of human escalation workflows

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Incident investigation. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of incident investigation

Reference example, not the only acceptable wording: Teacher explanation: Incident investigation preserves a timeline and distinguishes **observation** from hypothesis. Repair before **evidence** collection can erase clues.

Record timestamps, versions, requests, authorised actions and outcomes with minimised logs. Separate immediate containment from diagnosis. Compare candidate explanations against **evidence**.

Without opening the old lesson, explain one key idea from Release and rollback. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of release and rollback

Reference example, not the only acceptable wording: Teacher explanation: Release records bind code, configuration, data schema and evaluation **evidence**. Rollback requires compatibility with the current stored data.

Version v2 writes schema 2. Version v1 reads only schema 1. A rollback to v1 after v2 writes data cannot safely proceed without migration or a compatible reader.

Without opening the old lesson, explain one key idea from Caching and stale information. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of caching and stale information

Reference example, not the only acceptable wording: Teacher explanation: Caching reuses results but needs freshness and access-aware keys. A cached answer must not cross role or corpus-version boundaries.

Cache key includes query, role, corpus version and procedure version. Entry valid only while age<TTL. At exact TTL it expires. Teacher and student results never share a role-agnostic key.

Pictured end goal

Explain human escalation workflows using the deciding step in this fictional service.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Incident investigation

Release and rollback

Caching and stale information

Use the three-session teaching cycle above, with extra time as needed. Escalation routes unresolved decisions to an authorised person with sufficient context. It must define owner, deadline and fallback. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of human escalation workflows

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach a complete handoff without inventing delivery or approval

> Use accurate context, the correct owner and an explicit unresolved fallback.

Definitions and complete fictional handoff contract

Escalation routes unresolved decisions to an authorised person with sufficient context. It must define owner, deadline and fallback.

Toy workflow routes conflicting room approval to coordinator with source IDs and requested decision. While awaiting review the assistant returns needs_review and performs no booking.

Handoff element	Complete fictional classroom rule
Case and conflict	ROOM-C1: fictional source S1 says B12; S2 says D4
Requested decision	Resolve which room is approved; no assistant-selected booking
Responsible reviewer	Trusted coordinator role handles room_conflict
Evidence packet	Fictional case/source IDs, reported facts and unresolved question; minimum necessary context
Added local deadline	Virtual minute 30; exact equality is due, but never automatic approval
Fallback	Keep needs_review; no booking or certification
Picture route	Lesson-notes-only service routes reported equipment concern to assigned adult supervisor
Communication scope	Local packet/route only; no message sent and no human review claimed
Real-world scope	Fictional workflow; institutions define actual authorized responses and channels

Main state ledger and checked reasoning

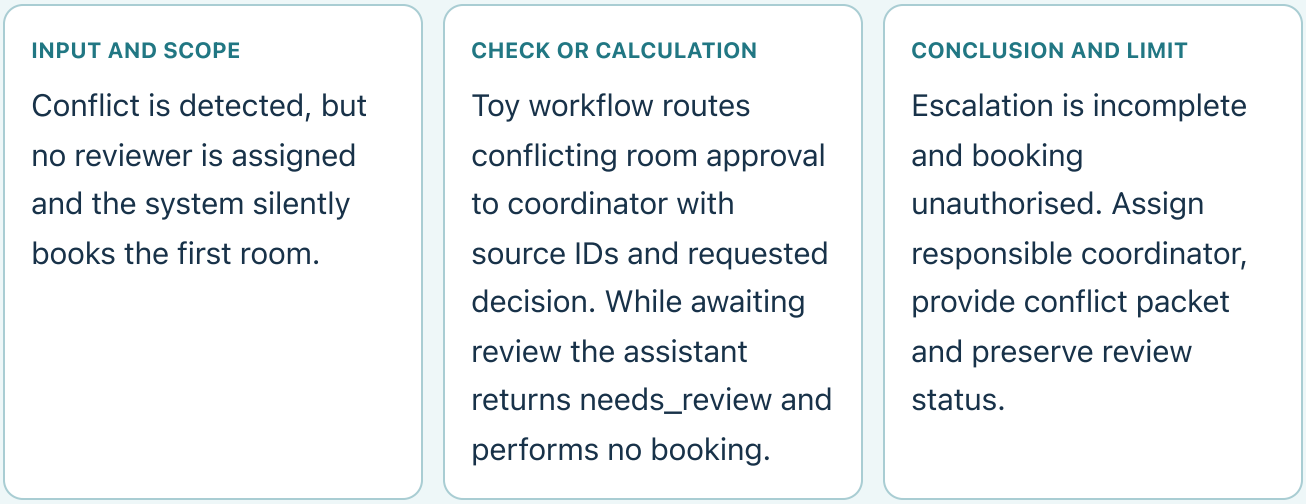
Main conflict state	Required handoff interpretation	Allowed action in model
No reviewer assigned	Escalation incomplete; choosing first room is unauthorized	Needs review; no booking

Main conflict state	Required handoff interpretation	Allowed action in model
Coordinator assigned with S1/S2 packet	Packet and route specified	Still needs review; no booking
Coordinator unreachable	Delivery/review not established	Keep unresolved; use only separately authorized fallback if configured
Local deadline reached without decision	Overdue, not approved	Needs review; no booking
Wrong reviewer role	Assignment does not match required authority	Keep unresolved
Handoff ready in code	Packet plus assigned/reachable symbolic route	No actual sending, receipt, decision or execution

Main conflict needs the responsible coordinator, S1/S2 evidence and a precise requested decision; no first-room booking is authorized. An assigned or reachable symbolic reviewer does not mean a real message was delivered or a decision received. Minute-30 equality makes the added local fixture overdue, still needs_review.

Picture lesson-notes bot cannot certify equipment: preserve the learner’s report as unverified, identify the designated supervisor route and keep unresolved status if unreachable. This is not an institutional laboratory procedure. Selected 16:00 case remains pending/review with no autoapproval; notification is allowed only through a separately authorized configured channel, not performed by this lesson code.

Human escalation workflows — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Escalation is incomplete and booking unauthorised. Assign responsible coordinator, provide conflict packet and preserve review status.

Qualification: Escalation without a responsible actor and declared fallback cannot guarantee resolution.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Conflict is detected, but no reviewer is assigned and the system silently books the first room.

Escalation is incomplete and booking unauthorised. Assign responsible coordinator, provide conflict packet and preserve review status.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s. The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain only authorized, minimized **evidence** without unnecessary private content so a later revision can be compared honestly.

Authority, timeout and communication probes

Escalation probe	Expected interpretation
Conflicting cards, no assigned reviewer	Incomplete route; do not pick first room
Assigned person lacks the relevant scope	Still no valid decision route
Packet contains only “please help”	Add case, source IDs, precise unresolved decision, owner and deadline
Reported concern recorded as confirmed diagnosis	Misstates evidence; preserve reported status
Reviewer not reachable	Keep unresolved; no invented contact or verdict
Deadline expires without reply	Pending/review persists under the declared no-autoapproval policy
Notification channel is absent or unauthorized	Do not claim or send a notification
Packet ready in a mock	Not proof of delivery, receipt, review or approval
Model highly confident	Confidence does not replace required authority

Separate reported equipment concern and service boundary

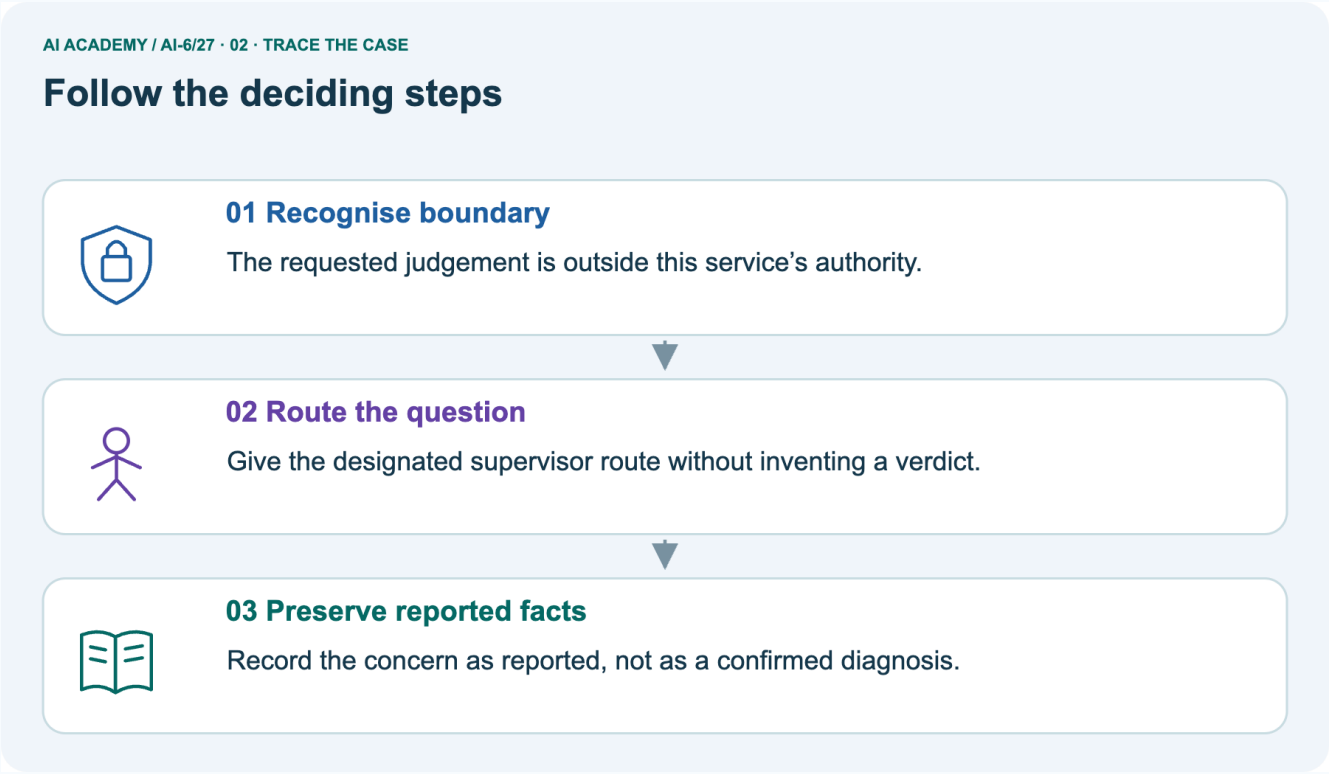
A useful human handoff. A learner asks whether a damaged laboratory bottle is safe; the bot handles lesson notes only.

The contract excludes equipment certification. An assigned adult supervisor handles equipment concerns.

The picture uses a different service boundary: a lesson-notes bot is asked to certify a reported damaged laboratory bottle. The packet must say “learner reports damage; physical condition not verified, ” rather than inventing a diagnosis or a safe/unsafe certificate. Route the concern to the designated adult supervisor under the fictional contract.

If nobody is reachable, the concern remains unresolved and the bot gains no inspection authority. A separately configured authorized fallback may be named, but neither contact nor approval should be invented. This example teaches authority and accurate handoff records, not a real laboratory response procedure.

Picture trace and unreachable-owner key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Recognise boundary	The requested judgement is outside this service’s authority.
2. Route the question	Give the designated supervisor route without inventing a verdict.
3. Preserve reported facts	Record the concern as reported, not as a confirmed diagnosis.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Escalation needs an appropriate person and accurate context. Model confidence cannot replace required authority or physical inspection.

This is a fictional workflow, not a real laboratory safety procedure; institutions define authorised responses.

Change: No supervisor is currently reachable.

Prediction: State unresolved status and use the institution’s approved waiting procedure.

Reason: Lack of a human reply does not expand the bot’s authority.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Escalation owner	The actor responsible for a reviewed decision
Decision packet	Evidence and exact action awaiting review
Fallback policy	Declared behaviour when review cannot complete

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

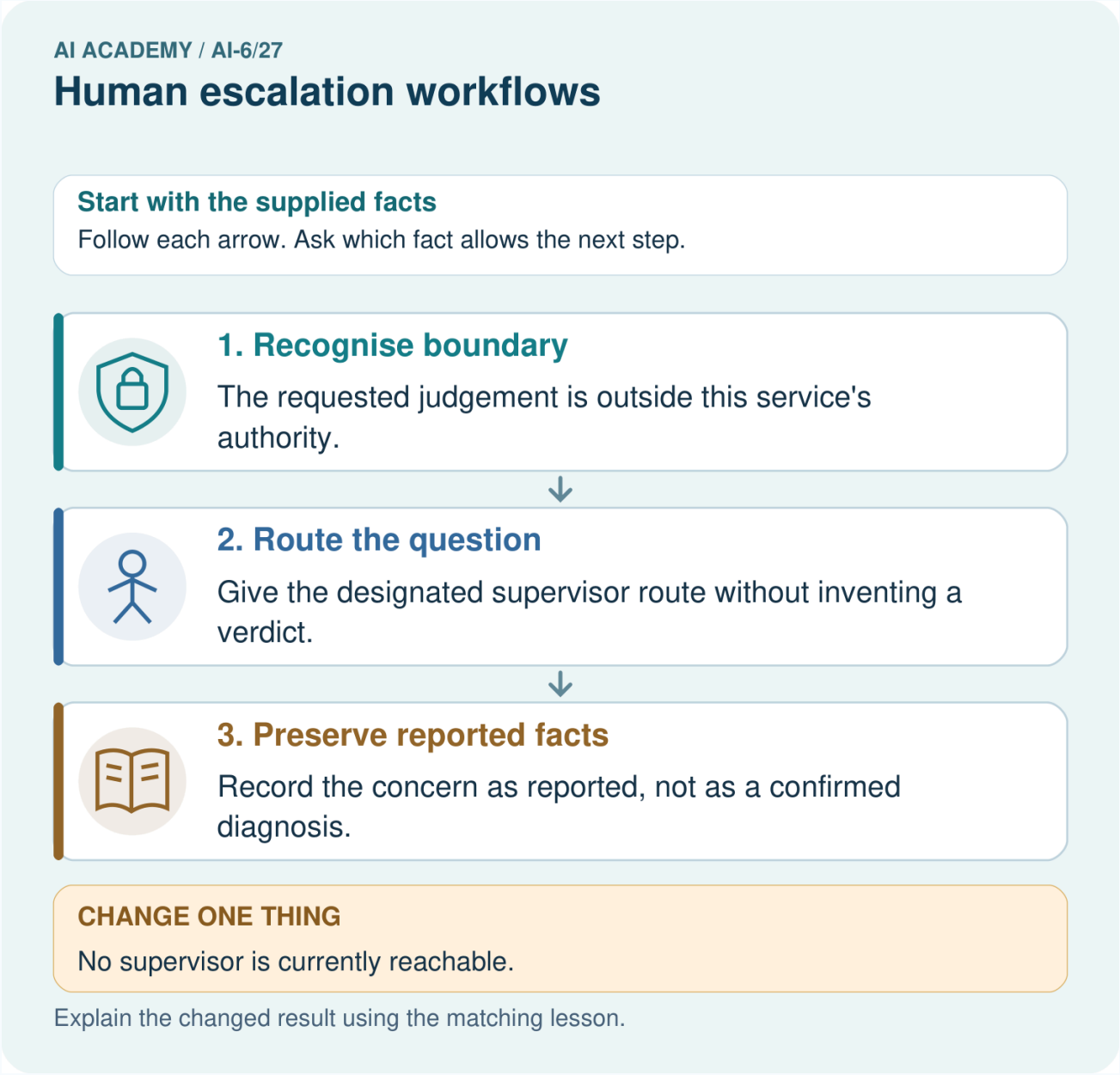
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Human escalation workflows

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Conflict is detected, but no reviewer is assigned and the system silently books the first room. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Escalation is incomplete and booking unauthorised. Assign responsible coordinator, provide conflict packet and preserve review status.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Human escalation workflows. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Escalation without a responsible actor and declared fallback cannot guarantee resolution.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Identify the exact service boundary in the bottle question. What accurate information belongs in the supervisor handoff, and what remains unresolved if nobody is reachable?

The requested judgement is outside this service's authority. Give the designated supervisor route without inventing a verdict. Record the concern as reported, not as a confirmed diagnosis.

Changed case: State unresolved status and use the institution's approved waiting procedure. Lack of a human reply does not expand the bot's authority.

Changed-case reasoning: State unresolved status and use the institution's approved waiting procedure. Lack of a human reply does not expand the bot's authority.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main conflict requires a scoped coordinator and accurate S1/S2 packet; assignment does not authorize booking. Picture unresolved reported equipment concern remains outside lesson-notes authority even if its supervisor is unreachable. Added local minute-30 deadline marks overdue without changing needs_review. Selected 16:00 case keeps pending/review because elapsed time is not approval.

Notify only through a separately authorized configured channel, and do not claim contact, receipt or approval without evidence.

Diagnose the current misconceptions

The worked result establishes every future case: Escalation without a responsible actor and declared fallback cannot guarantee resolution.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Reviewer deadline is 16:00; no reply by 16:00. Policy says remain pending, not auto-approve. Decide.

Teacher explanation: Keep pending/review outcome; elapsed time is not approval. Notify through only the separately authorised channel if configured.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Escalation routes unresolved decisions to an authorised person with sufficient context. It must define owner, deadline and fallback.

Toy workflow routes conflicting room approval to coordinator with source IDs and requested decision. While awaiting review the assistant returns needs_review and performs no booking.

W2 Trace the worked case

Conflict is detected, but no reviewer is assigned and the system silently books the first room. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Escalation is incomplete and booking unauthorised. Assign responsible coordinator, provide conflict packet and preserve review status.

W3 Audit the limit

A report omits this qualification: Escalation without a responsible actor and declared fallback cannot guarantee resolution. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Escalation without a responsible actor and declared fallback cannot guarantee resolution.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Escalation without a responsible actor and declared fallback cannot guarantee resolution.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Escalation is incomplete and booking unauthorised. Assign responsible coordinator, provide conflict packet and preserve review status.

Qualification: Escalation without a responsible actor and declared fallback cannot guarantee resolution.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Reviewer deadline is 16:00; no reply by 16:00. Policy says remain pending, not auto-approve. Decide.

Teacher explanation: Keep pending/review outcome; elapsed time is not approval. Notify through only the separately authorised channel if configured.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Keep pending/review outcome; elapsed time is not approval. Notify through only the separately authorised channel if configured. Require intermediate reasoning and the stated scope.

Fresh retry: Review deadline passes with no response; fallback explicitly denies action. Expected reasoning: Deny action according to policy; do not infer approval from silence.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

Add a labelled human escalation workflows **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of human escalation workflows.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Accessibility and user feedback. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/28, ai-6/30, ai-6/33. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of human escalation workflows

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Session:** Session 3 · Investigate and transfer · 45 min
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>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Conflict is detected, but no reviewer is assigned and the system silently books the first room. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Escalation is incomplete and booking unauthorised. Assign responsible coordinator, provide conflict packet and preserve review status.

Check every learner before discussion: Use the worked case for Human escalation workflows. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Escalation without a responsible actor and declared fallback cannot guarantee resolution.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: No supervisor is currently reachable. Require a **prediction** and a reason before comparing. Expected result: State unresolved status and use the institution’s approved waiting procedure.

Reason: Lack of a human reply does not expand the bot’s authority.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

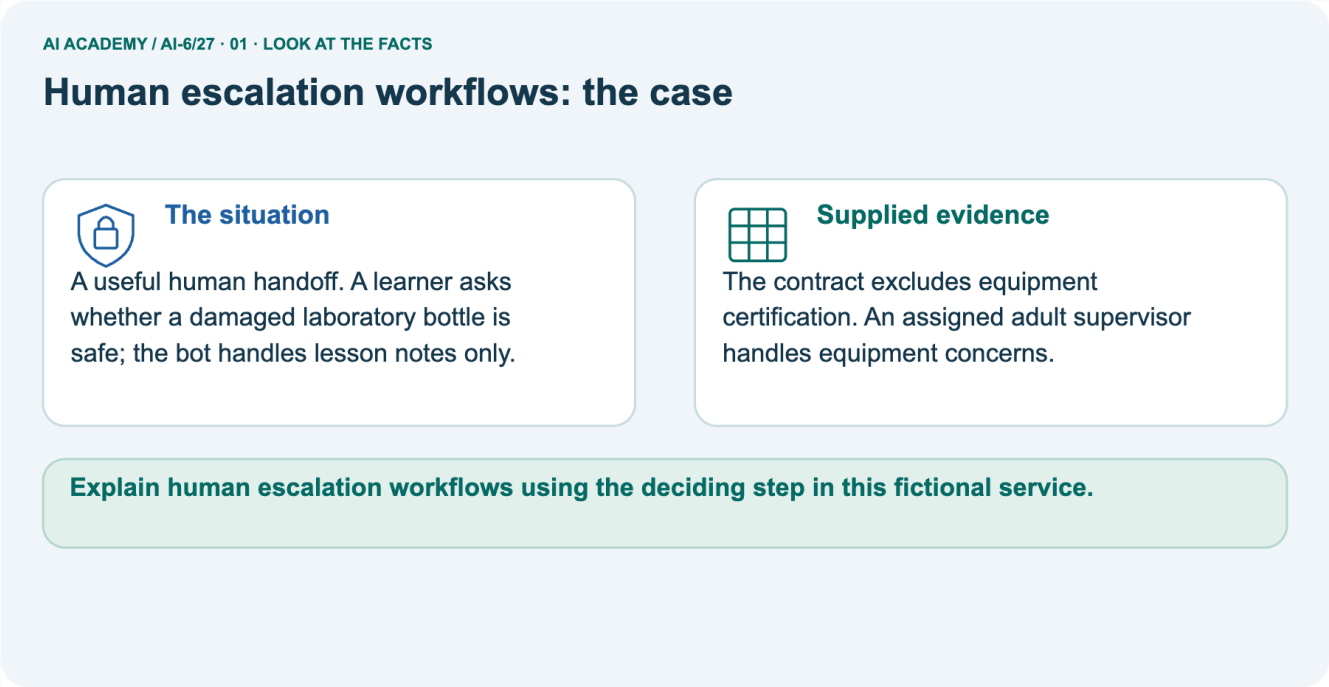
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Human escalation workflows: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

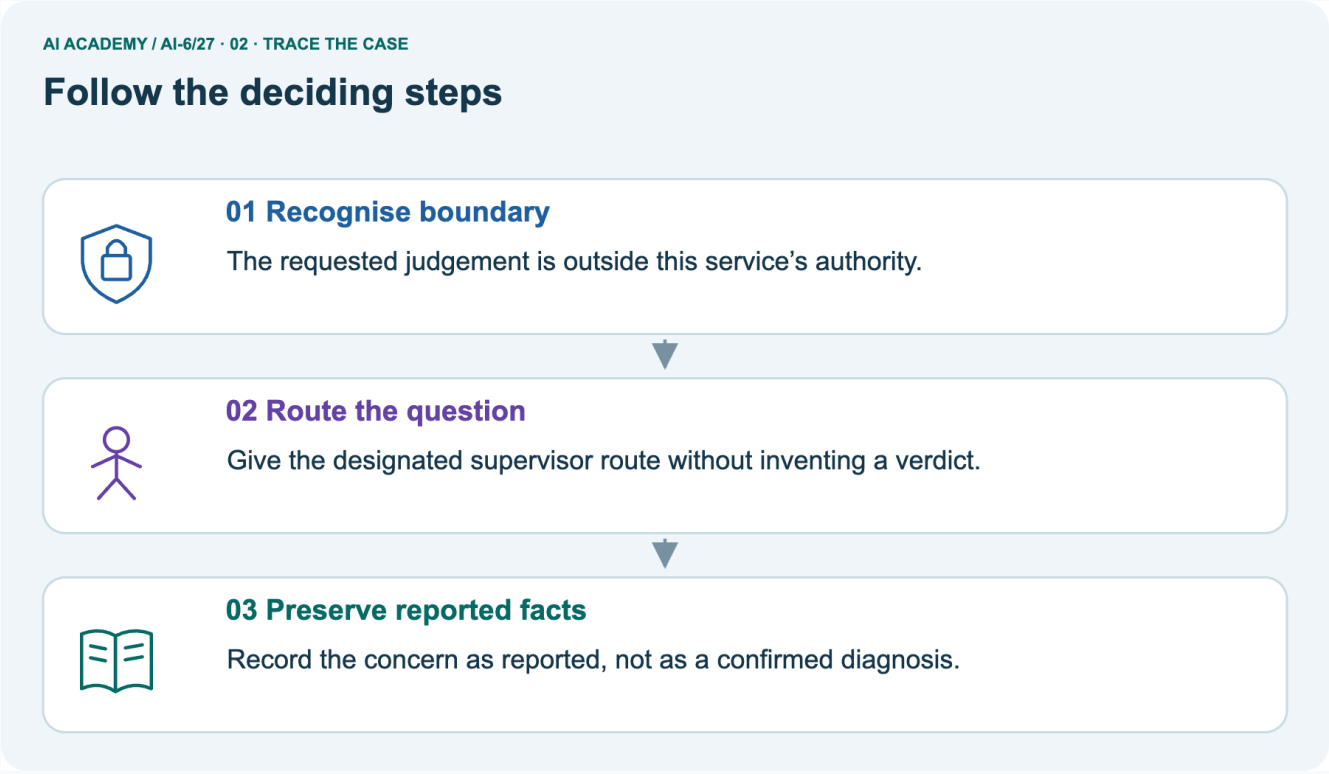
A useful human handoff. A learner asks whether a damaged laboratory bottle is safe; the bot handles lesson notes only. Supplied facts: The contract excludes equipment certification.

An assigned adult supervisor handles equipment concerns.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



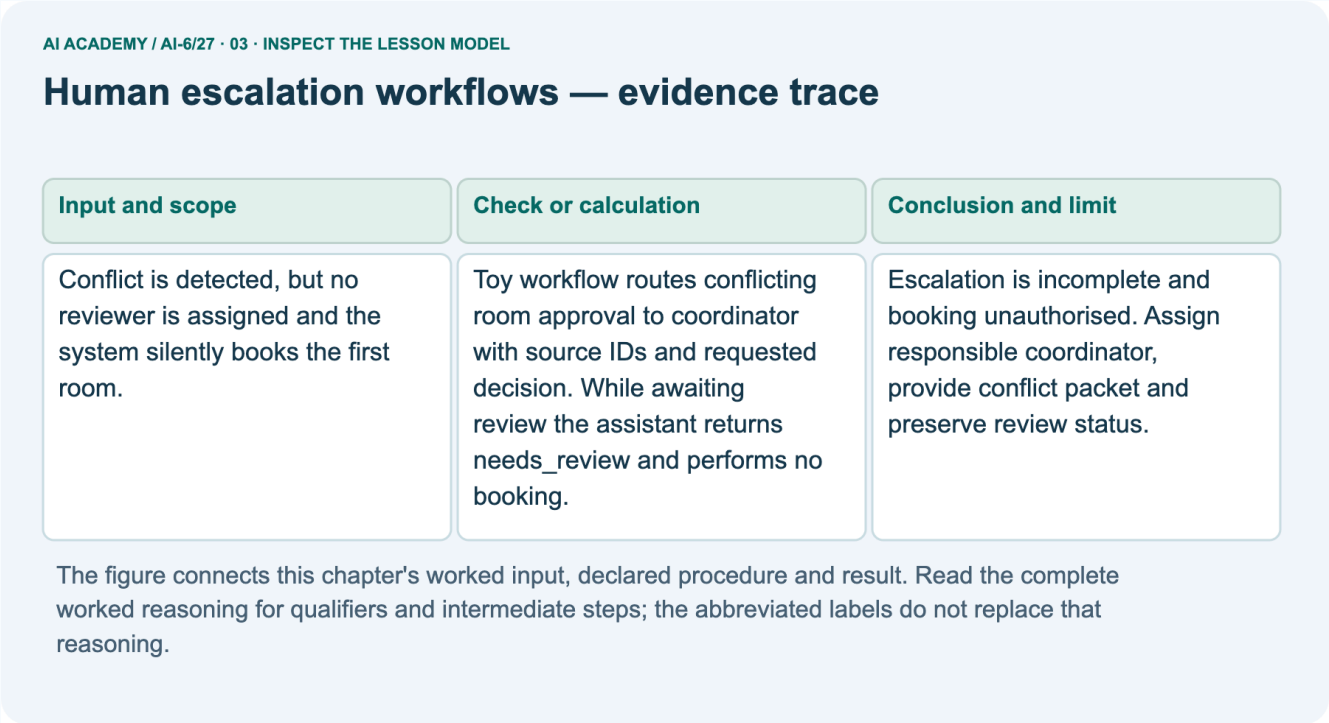
Follow the labelled picture with the matching explanation. Open the vector diagram

Recognise boundary: The requested judgement is outside this service's authority. → Route the question: Give the designated supervisor route without inventing a verdict. → Preserve reported facts: Record the concern as reported, not as a confirmed diagnosis.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Human escalation workflows — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Human escalation workflows — **evidence** trace.

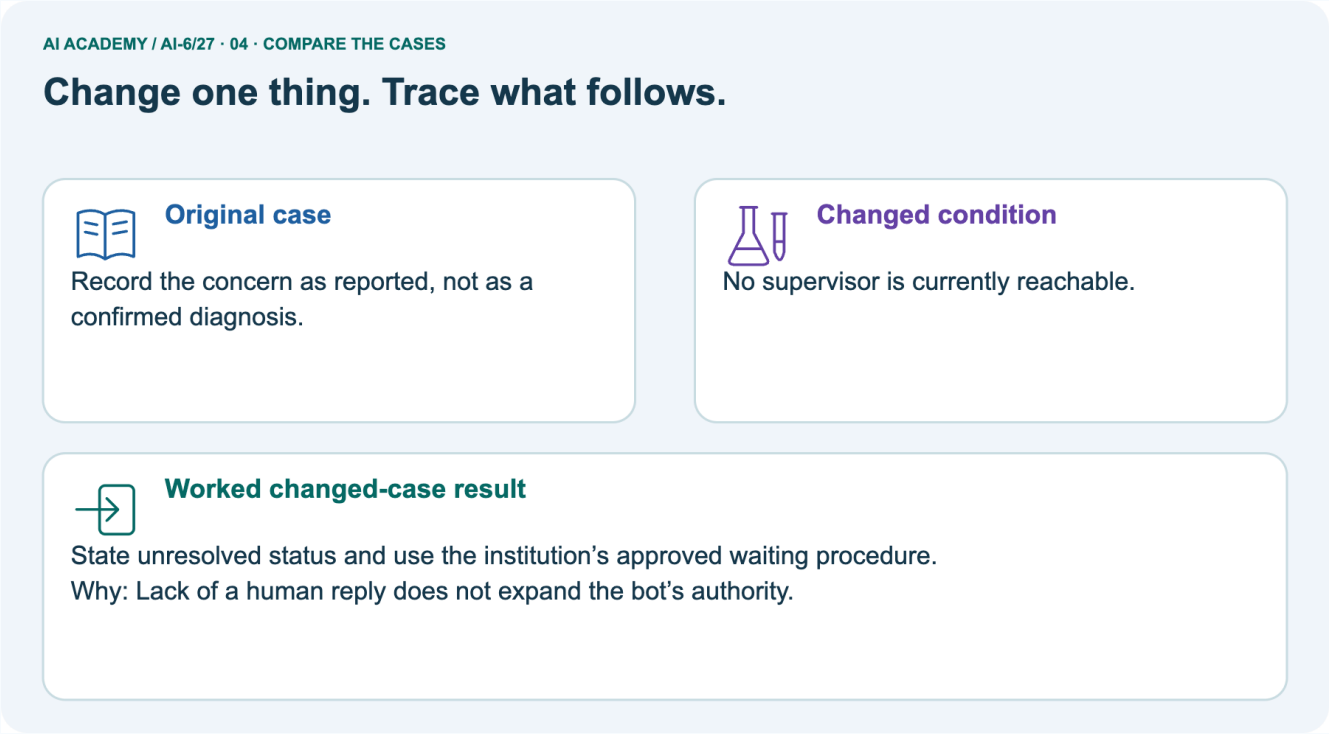
Conflict is detected, but no reviewer is assigned and the system silently books the first room. | Toy workflow routes conflicting room approval to coordinator with source IDs and requested decision.

While awaiting review the assistant returns needs_review and performs no booking. | Escalation is incomplete and booking unauthorised. Assign responsible coordinator, provide conflict packet and preserve review status.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Record the concern as reported, not as a confirmed diagnosis. Changed condition: No supervisor is currently reachable.

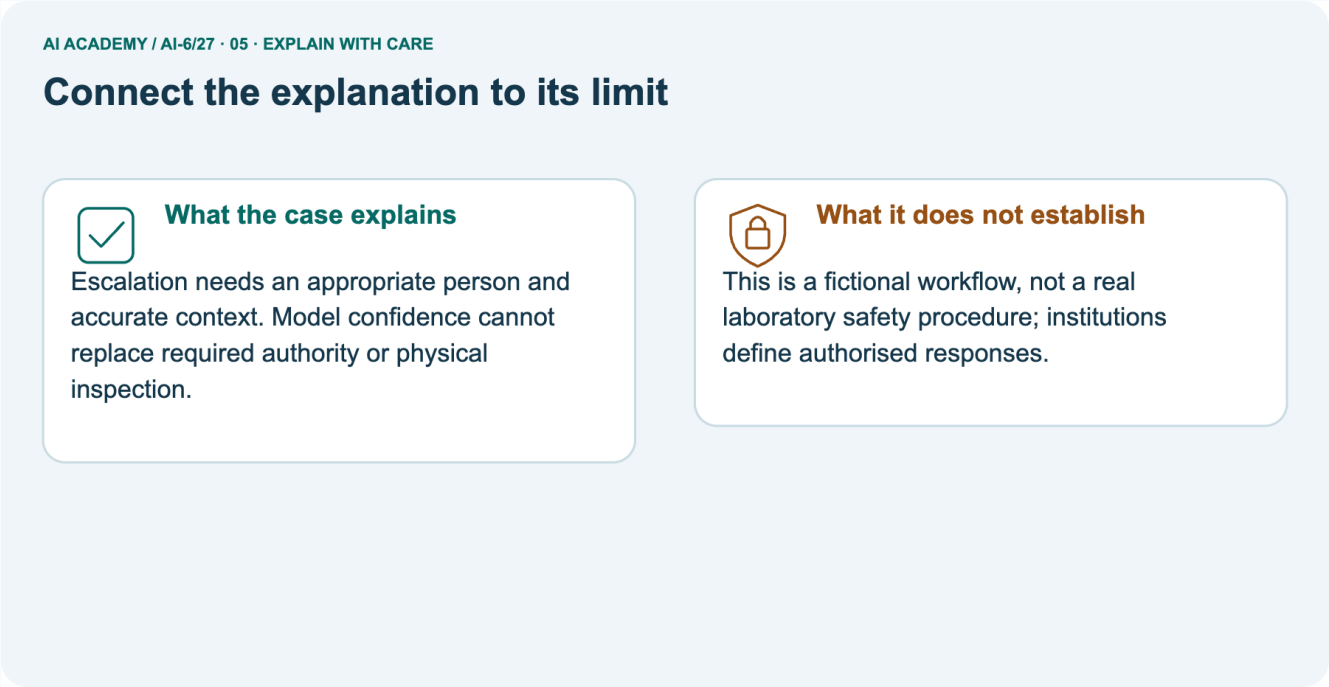
Expected result: State unresolved status and use the institution’s approved waiting procedure. Reason: Lack of a human reply does not expand the bot’s authority.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Lack of a human reply does not expand the bot’s authority.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Escalation needs an appropriate person and accurate context. Model confidence cannot replace required authority or physical inspection.

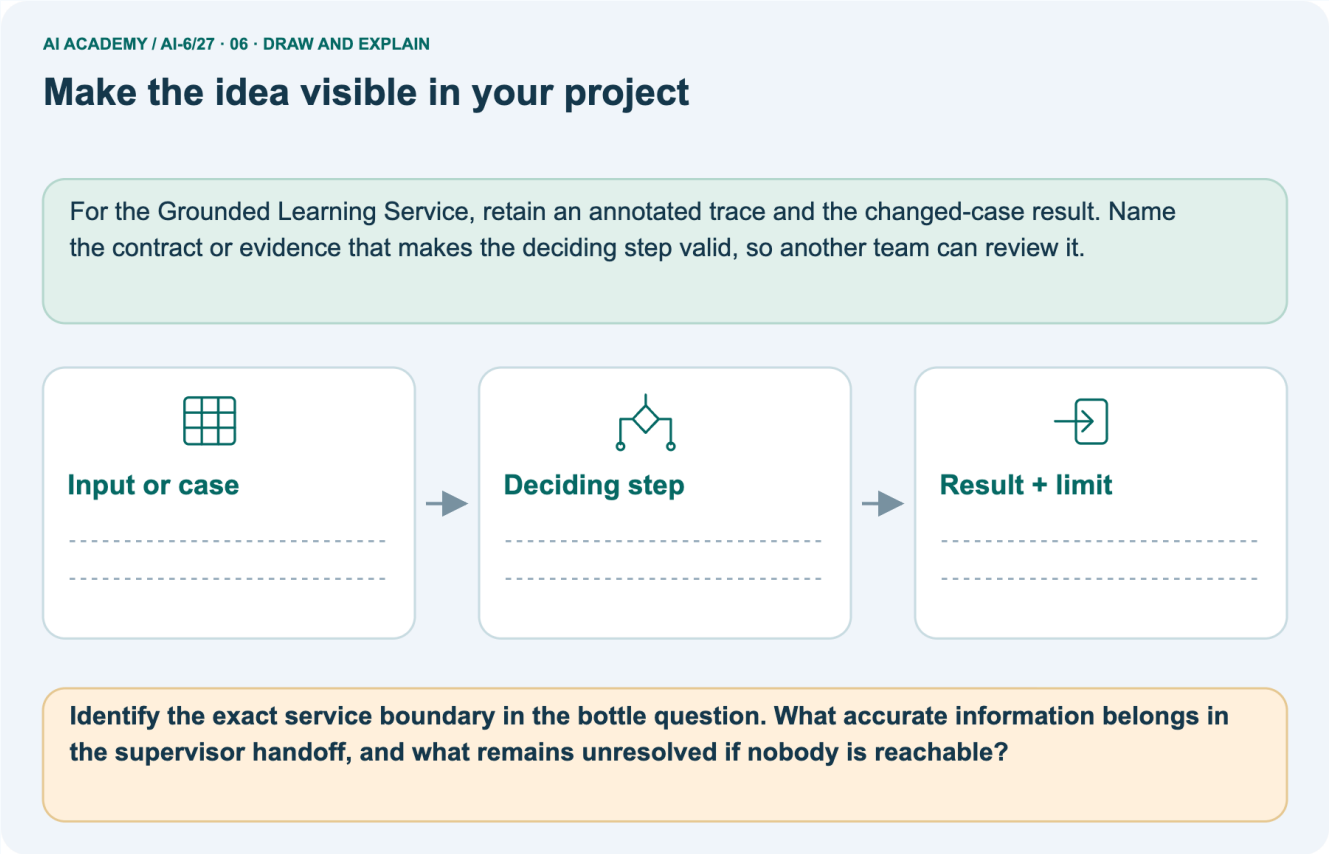
Limit: This is a fictional workflow, not a real laboratory safety procedure; institutions define authorised responses.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This is a fictional workflow, not a real laboratory safety procedure; institutions define authorised responses.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Identify the exact service boundary in the bottle question.

What accurate information belongs in the supervisor handoff, and what remains unresolved if nobody is reachable?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The requested judgement is outside this service’s authority. Give the designated supervisor route without inventing a verdict. Record the concern as reported, not as a confirmed diagnosis.

Changed case: State unresolved status and use the institution’s approved waiting procedure. Lack of a human reply does not expand the bot’s authority.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Human escalation workflows: the case

02. Follow the deciding steps

03. Human escalation workflows — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.